



**Informatique et systèmes
de communication ISC**



Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften

Signal and information (SigInf)

204.1

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Lesson 5. Image processing

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1. INTRODUCTION TO IMAGES

An image is a 2D signal

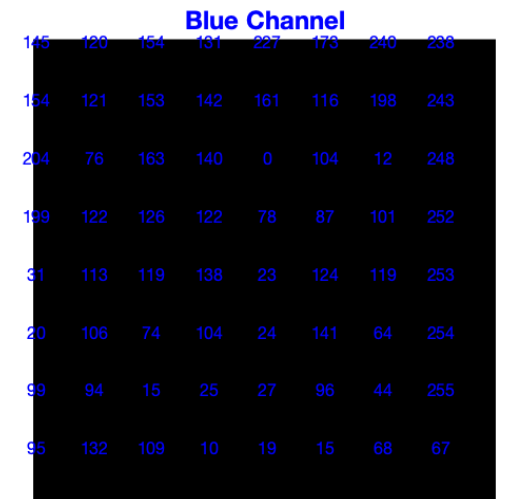
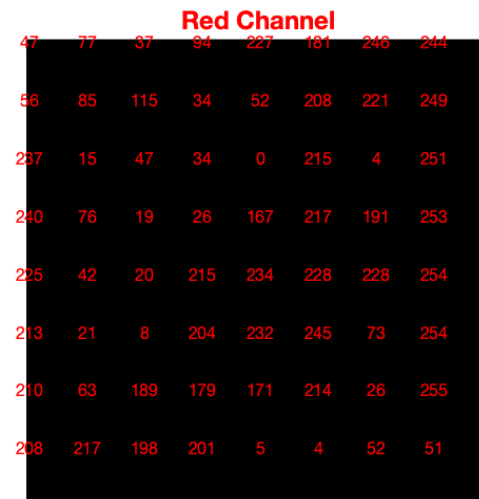
An image is a 2D grid of pixel values, each representing light intensity. Greyscale images are single-channel arrays and color images are three-channel (RGB) arrays.

Definition

An image is a 2D signal: greyscale



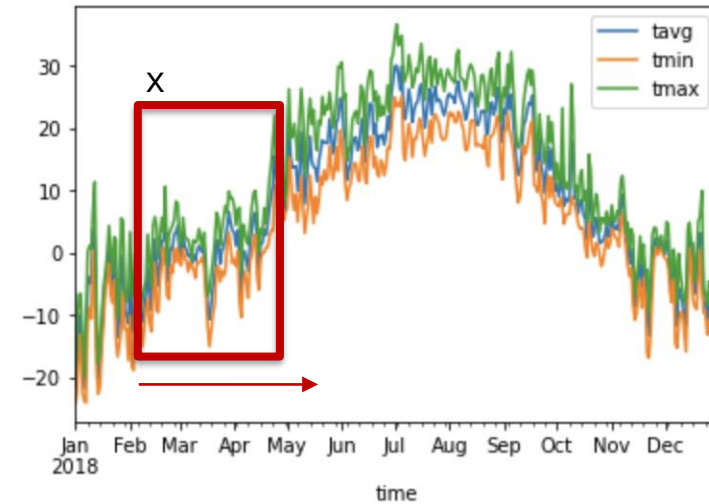
An image is a 2D signal: RGB image



An image is a 2D signal



We investigate spatial relationships between pixels



(where here your x is time)

2. NOISE

Noise in an image

Noise: as for 1D signal, noise is a variation that perturbs the signal clarity (or message that the image conveys), so it makes the interpretation harder.

Filtering: clean the noise.

Original Grayscale Image



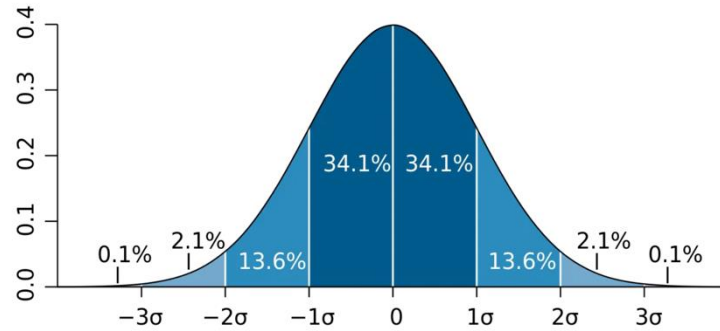
With Gaussian Noise



With Salt-and-Pepper Noise



Gaussian Noise in an image



Noise: to all pixel values, random noise sample from a Gaussian distribution is added – so that it appears as random (but smooth) variations across the image.

Original Grayscale Image



With Gaussian Noise



<https://wiki.cloudfactory.com/docs/mp-wiki/augmentations/gaussian-noise>

Salt and Pepper Noise in an image

Noise: randomly changes some pixels to either the maximum intensity (salt) or the minimum intensity (pepper) – this is caused by data transmission errors, for example.

Original Grayscale Image



With Salt-and-Pepper Noise



3. FILTERING IN THE SPATIAL DOMAIN

Image preprocessing techniques in the spatial domain

Filtering: we apply filtering to manipulate pixel values in an image, in order to make certain features appearing, such as edges, or to erase noise.

Category	Technique	Purpose	Examples
Smoothing	Mean Filter	Reduces Gaussian noise, smooths image.	Applies uniform averaging to pixel neighborhoods, but blurs edges.
	Gaussian Filter	Reduces noise while preserving edges.	Uses a Gaussian kernel for weighted averaging (smooths without excessive blurring).
	Median Filter	Removes salt-and-pepper noise, preserves edges.	Replaces pixel with the median value of its neighbors.
	Bilateral Filter	Smooths image while keeping edges sharp.	Combines spatial and intensity weighting to prevent edge blurring.
Sharpening	Laplacian Filter	Highlights edges and fine details by enhancing high frequencies.	Computes second-order derivatives to detect intensity changes.
	Unsharp Masking	Enhances details while keeping the image natural.	Subtracts a blurred image from the original, amplifies differences, and adds them back.
	Gradient Filters	Detects edges (intensity gradients).	Sobel, Prewitt operators calculate gradients in horizontal or vertical directions.
Noise Reduction	Gaussian Filter	Reduces random noise.	Blurs high-frequency noise using a Gaussian kernel.
	Median Filter	Removes noise while preserving edges.	Excellent for removing salt-and-pepper noise.
	Bilateral Filter	Reduces noise, preserves edges.	Suitable for applications like medical imaging or facial feature smoothing.
Contrast Enhancement	Histogram Equalization	Enhances global contrast.	Redistributes pixel intensities to span the full intensity range.
	Adaptive Histogram Equalization (AHE)	Improves local contrast, enhances details in varying lighting.	Divides the image into smaller regions (tiles) for local intensity redistribution (e.g., CLAHE).

Smoothing filtering (low-pass filtering): mean filtering

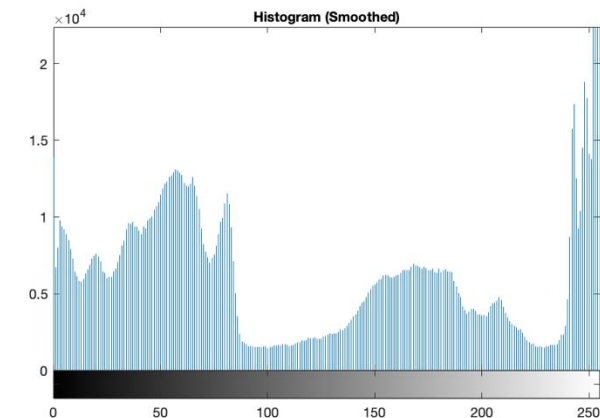
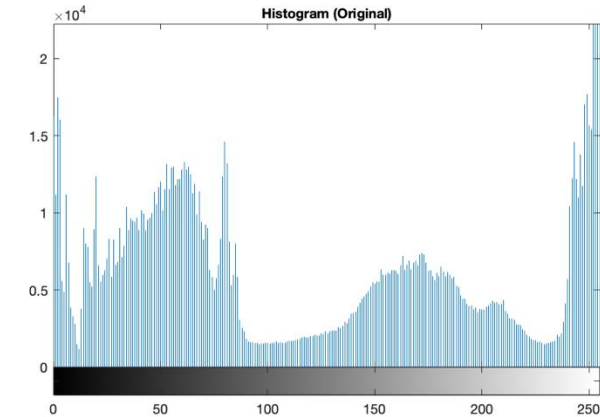
Smoothing: Smoothing reduces noise and removes sharp intensity variations by averaging pixel intensities.

Mean filtering:

- Replaces each pixel with the average of its neighbourhood.
- **Kernel Example:**

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

- **Use Case:** Reduces Gaussian noise but blurs edges.



Smoothing filtering (low-pass filtering): Gaussian filtering. It is also a noise reduction filtering

Smoothing: Smoothing reduces noise and removes sharp intensity variations by averaging pixel intensities.

Gaussian filtering:

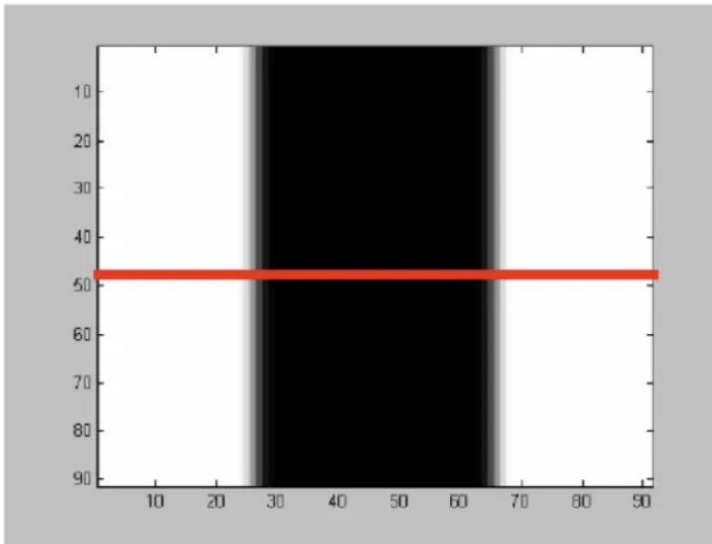
- Uses a Gaussian-shaped kernel for weighted averaging, preserving edges better than the mean filter
- **Use Case:** Preserves edges and can also remove the noise.



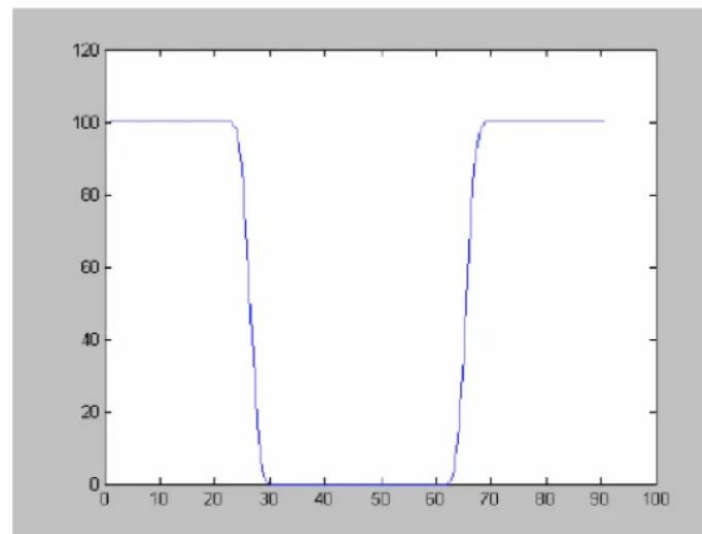
Sharpening filtering (high-pass filtering): Gradient

Sharpening filters (gradient): detect edges by computing intensity gradients along image (x,y) direction (or a combination of both).

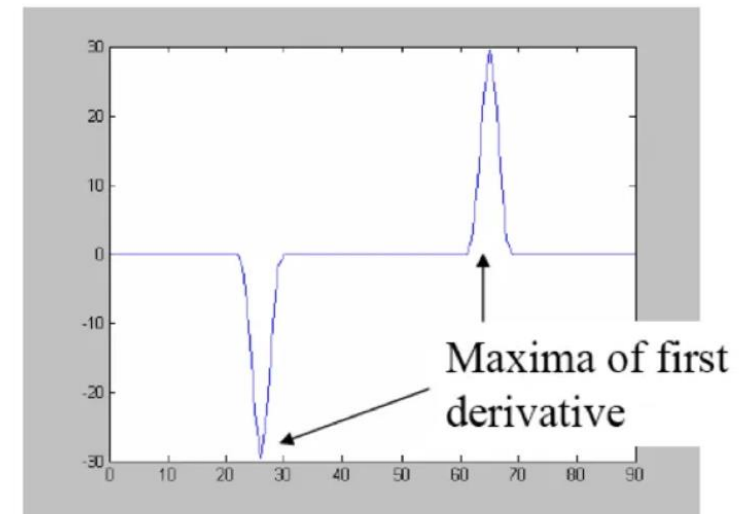
Used for Edges detection (or features detection)



Image



Intensities of pixels on the red line



1st derivative of pixel intensities

<https://medium.com/jun94-devpblog/cv-3-gradient-and-laplacian-filter-difference-of-gaussians-dog-7c22e4a9d6cc>

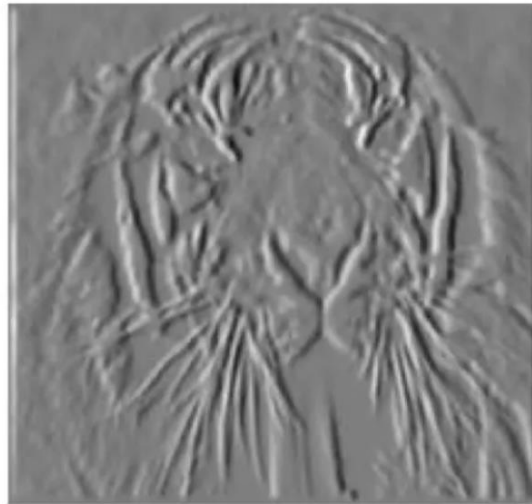
Sharpening filtering (high-pass filtering): Gradient

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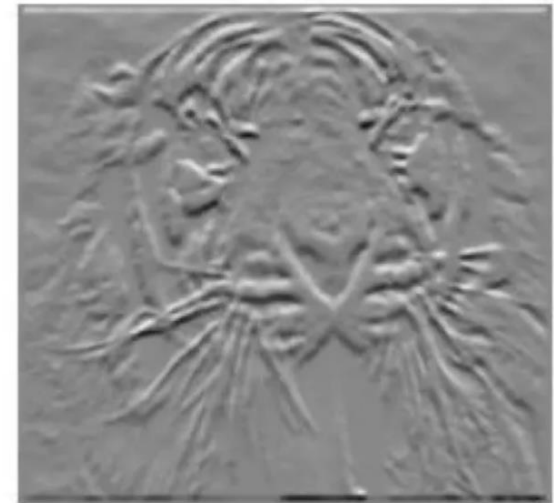
Used for Edges detection (or features detection)



Image f



$$\frac{\partial f(x, y)}{\partial x}$$



$$\frac{\partial f(x, y)}{\partial y}$$

Sharpening filtering (high-pass filtering): Laplacian

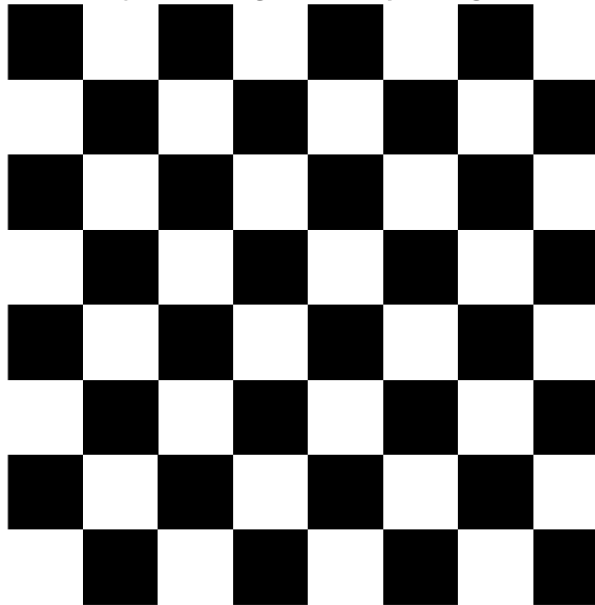
Sharpening filters (Laplacian):

another edge detector, it highlights regions where pixels intensity dramatically change.

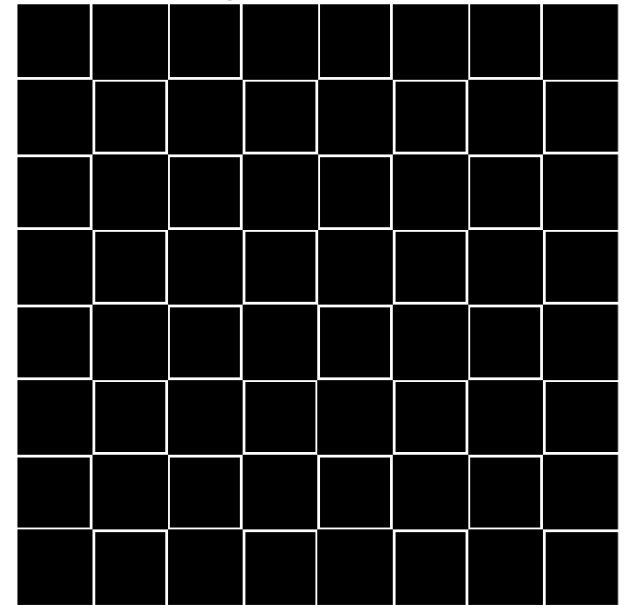
Matrix kernel:

0	-1	0
-1	4	-1
0	-1	0

Synthetic Image with Abrupt Changes

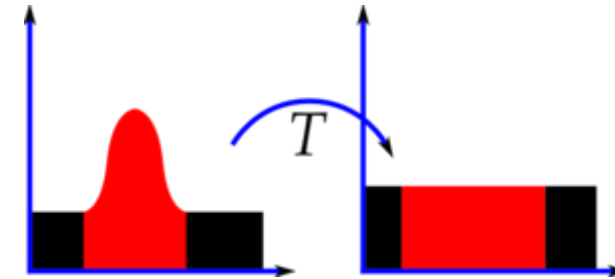


Laplacian Filter Result



Contrast enhancement via histogram equalization

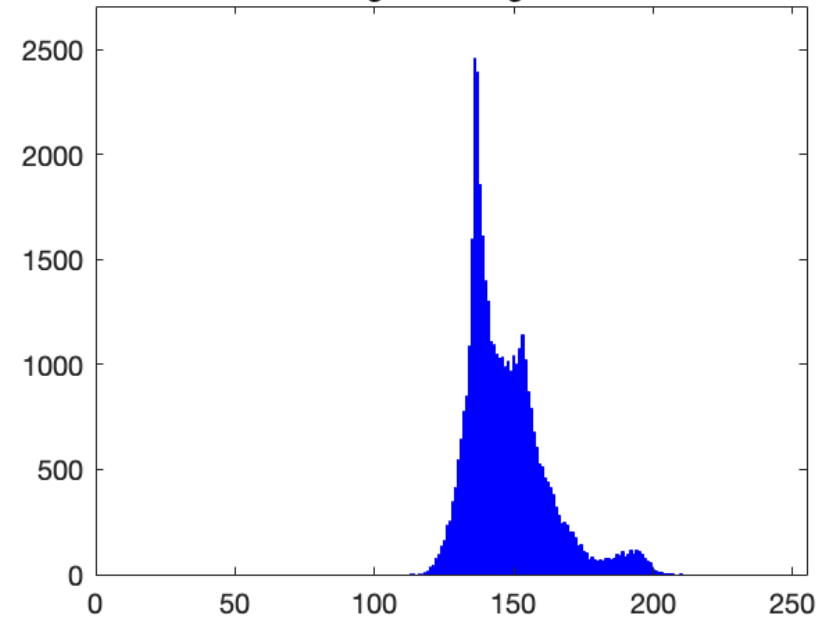
Enhancement: we redistribute the pixel intensity over the entire available range. To do so, we map the cumulative distribution of the intensity values to a linear distribution. **This enhances (increase the contrast) of the image.**



Original Image



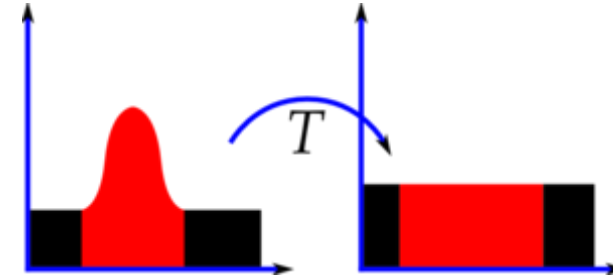
Original Histogram



https://docs.opencv.org/4.x/d5/daf/tutorial_py_histogram_equalization.html

Image preprocessing techniques: enhancement

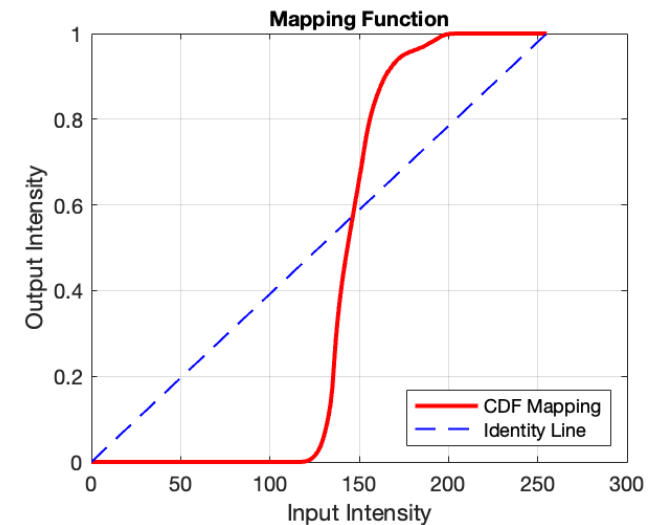
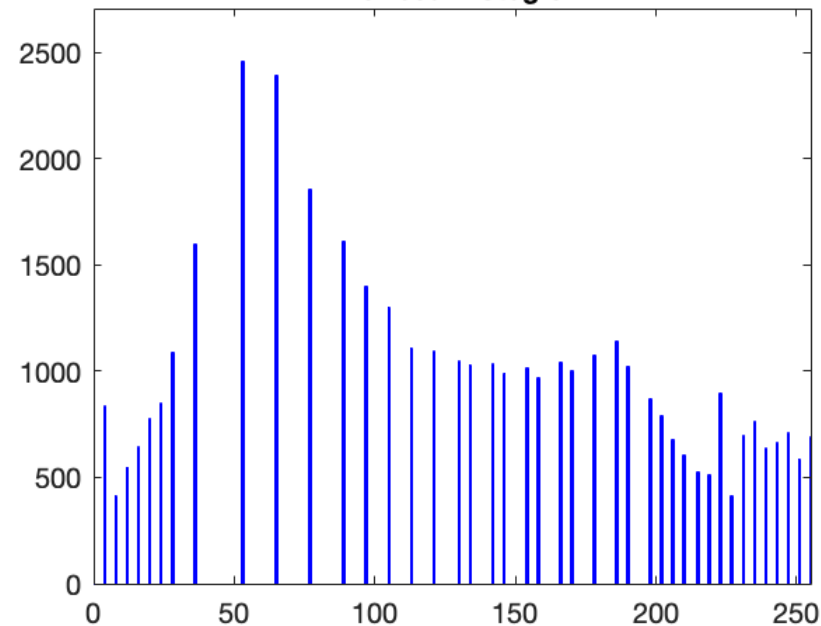
Enhancement: we redistribute the pixel intensity over the entire available range. To do so, we map the cumulative distribution of the intensity values to a linear distribution. **This enhances (increase the contrast) of the image.**



Enhanced Image



Enhanced Histogram



https://docs.opencv.org/4.x/d5/daf/tutorial_py_histogram_equalization.html

4. FILTERING IN THE FREQUENCY DOMAIN

Frequency domain analysis: recap FT on images

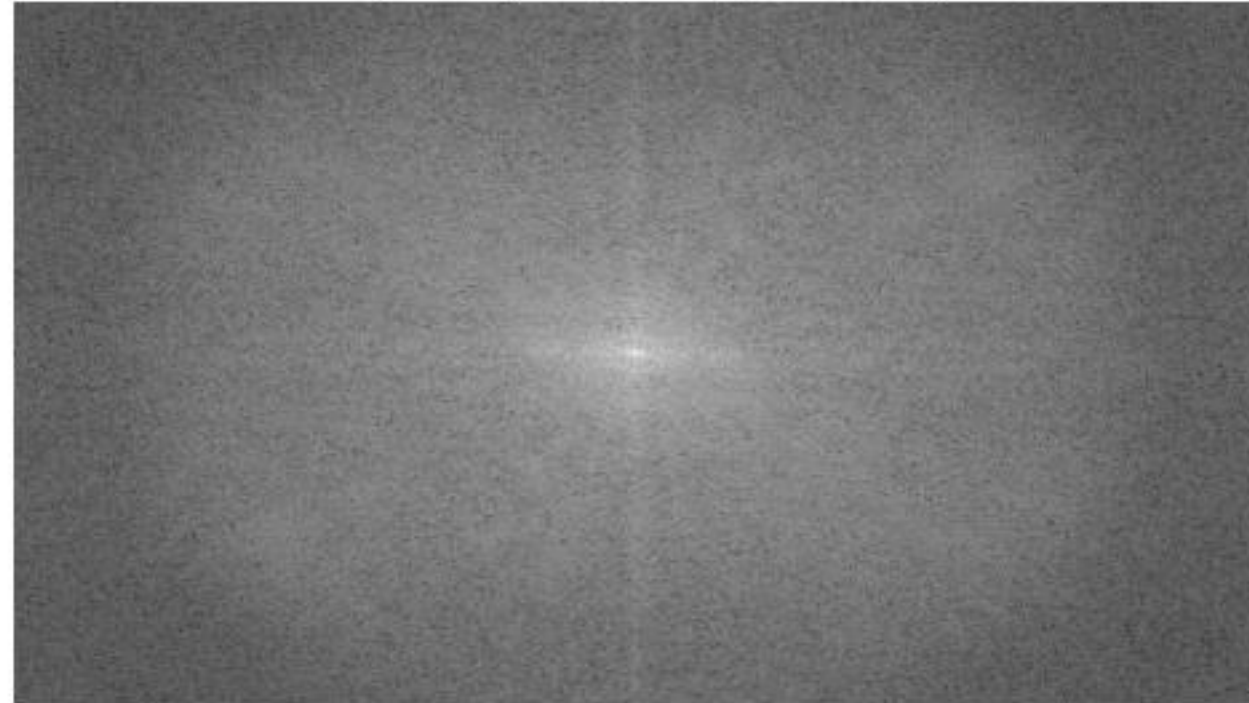
Low frequencies: responsible for the slowly changing patterns, such as the average values and smooth transitions.

Higher (increasing) frequencies: sharp changes (edges)

Original Image

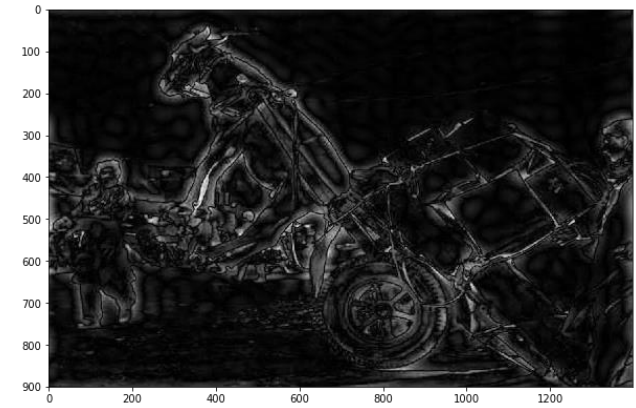
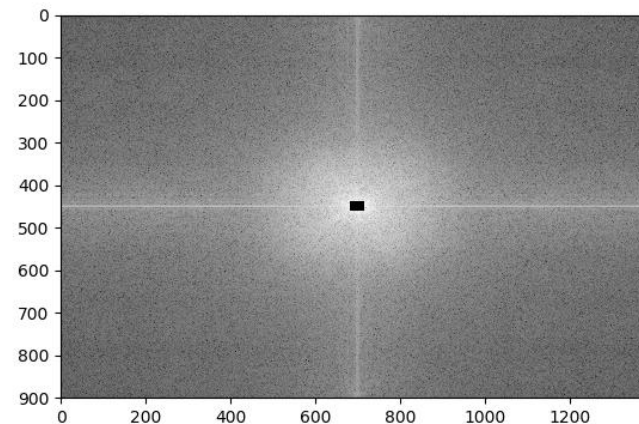
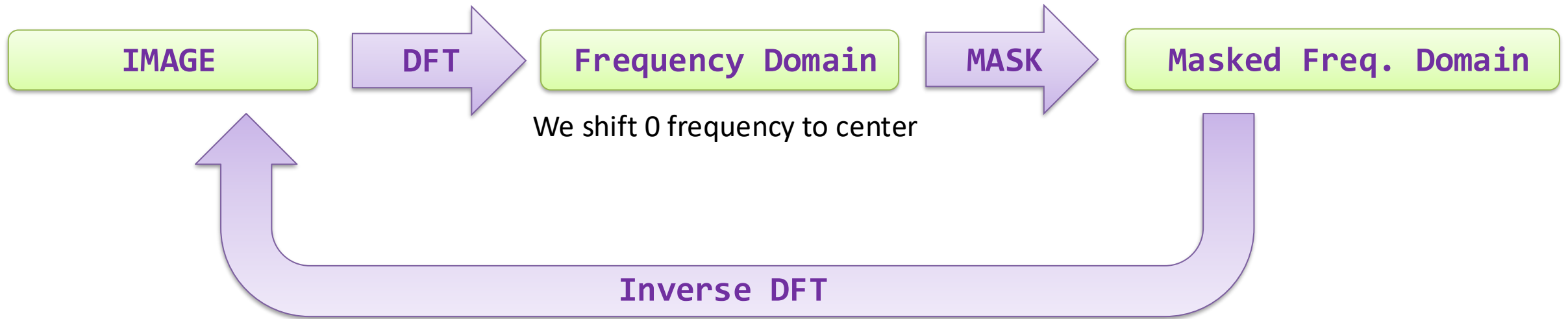


K-space (Frequency Domain)



[Go back on the DFT tutorial from Mauro!](#)

Low-pass and high-pass filtering (here high-pass)



Code break (20 minutes)

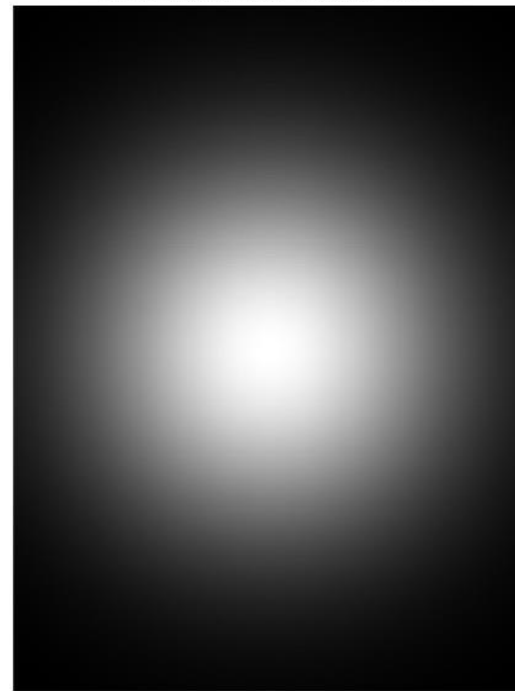
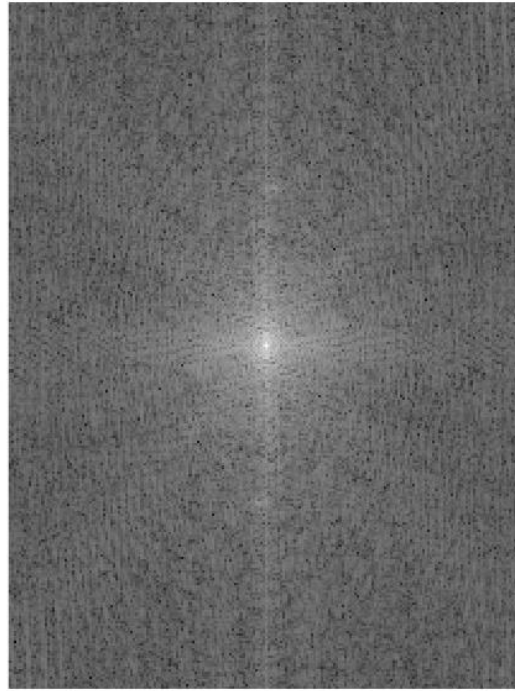
Implement:

- A low-pass filtering in the previous image
- A Gaussian shaped filter that attenuates higher frequency (and that allows low frequency to pass)
- A band-pass filter: meaning we allow frequencies within a specific range to pass

Code break (20 minutes)

Implement:

- A Gaussian shaped filter that attenuates higher frequency (and that allows low frequency to pass)

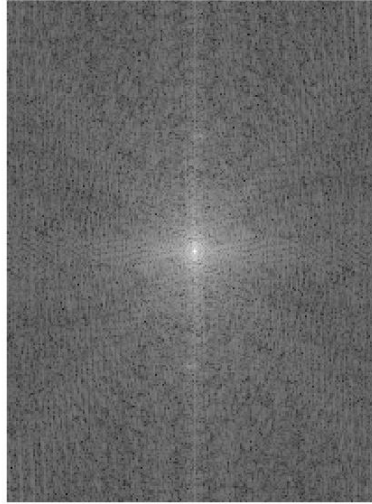


Implement: a band-pass filter: meaning we allow frequencies within a specific range to pass

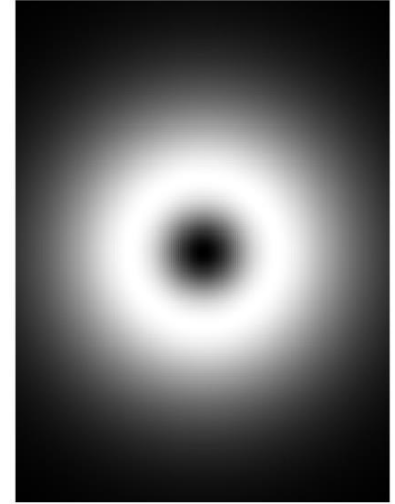
Original Image



Magnitude Spectrum (Original)



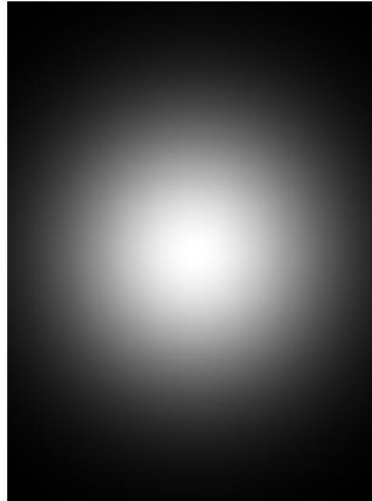
Band-Pass Filter



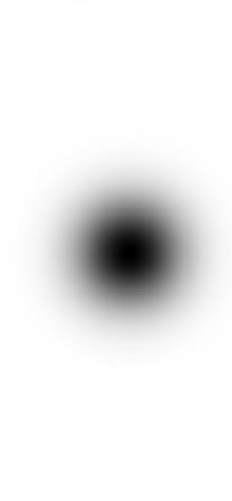
Filtered Image (Band-Pass)



Low-Pass Filter



High-Pass Filter



Spatial vs. Frequency domain processing comparison

Aspect	Spatial Domain	Frequency Domain
Operation	Local filtering with kernels (e.g., convolution).	Global filtering of frequency components.
Focus	Works on pixel intensities.	Works on frequency representation of the image.
Applications	Smoothing, sharpening, edge detection.	Noise reduction, smoothing, periodic pattern removal.
Advantages	Easier to implement and interpret.	Allows precise control over global features.
Limitations	Ineffective for global patterns or periodic noise.	Computationally intensive for large images.

5. FEATURES EXTRACTION

What does it mean

Features extraction: the process of identifying and extracting relevant information (features) from an image to facilitate understanding, analysis, or classification. Features are specific attributes or patterns within the image, such as edges, corners, or textures.

Example: Edge detection as feature detection (**Canny – popular filter**)

Original Grayscale Image

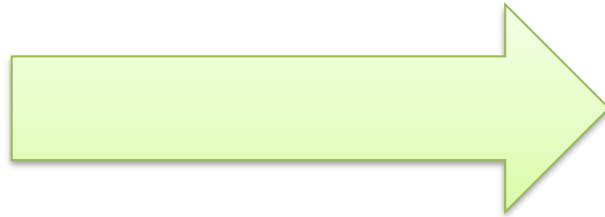


Canny Edge Detection



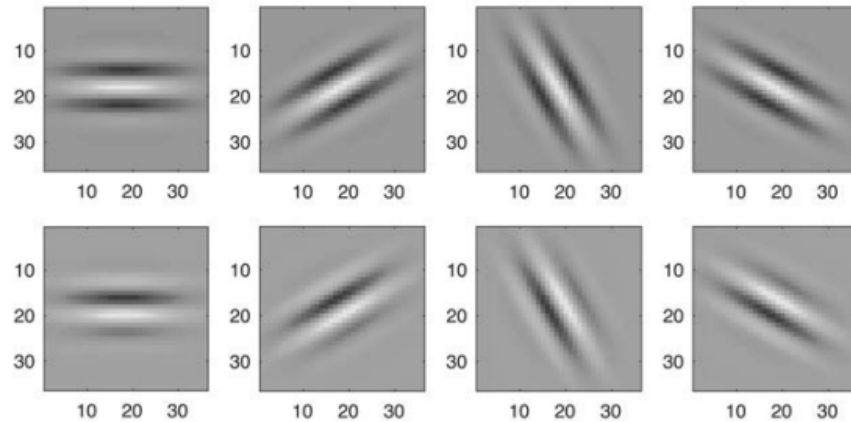
Segmentation

Jaime's tutorial



Gabor filters: extracting orientation

Gabor filter code: [link](#)

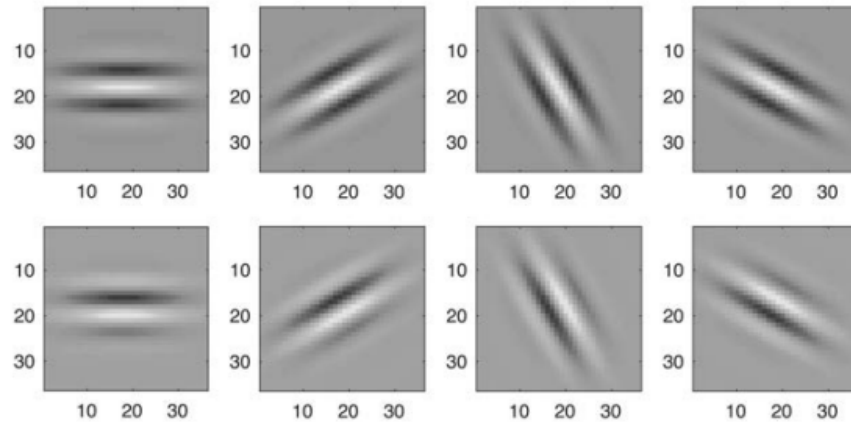


In each image: (top) even part of Gabor filters (real part), (bottom) odd one (imaginary part). Corresponding orientation from left to right: $\theta = 0$, $\theta = \pi/6$, $\theta = 2\pi/3$, $\theta = 5\pi/6$, with $\sigma = 4.48$ pixels



Gabor filters: extracting orientation

Gabor filter code: [link](#)



In each image: (top) even part of Gabor filters (real part), (bottom) odd one (imaginary part). Corresponding orientation from left to right: $\theta = 0$, $\theta = \pi/6$, $\theta = 2\pi/3$, $\theta = 5\pi/6$, with $\sigma = 4.48$ pixels

